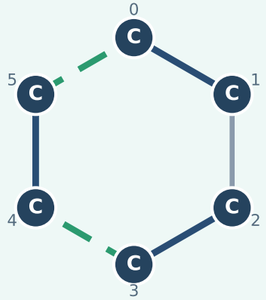
Pericyclic transition-state orbital circuits

Eight formally thermally allowed approaches from the dataset templates • Hückel: 4k+2 e⁻ • Möbius: 4k e⁻

π bond · conjugated link · forming bond · breaking bond · one orbital phase inversion

Diels–Alder [4+2]

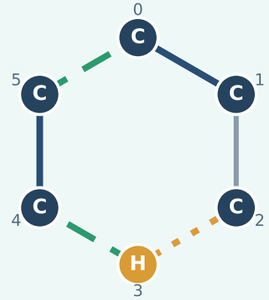
6 e⁻ • Hückel • 4π + 2π



suprafacial / suprafacial — **THERMAL ALLOWED**

Alder–ene

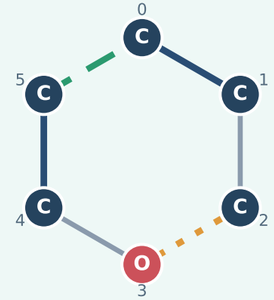
6 e⁻ • Hückel • 2π + 2π + C-H σ



all suprafacial — **THERMAL ALLOWED**

Claisen [3,3]

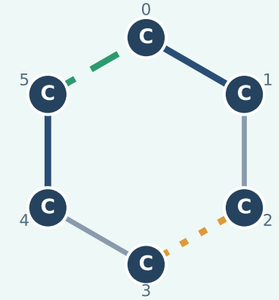
6 e⁻ • Hückel • 2π + 2π + C-O σ



suprafacial / suprafacial — **THERMAL ALLOWED**

Cope [3,3]

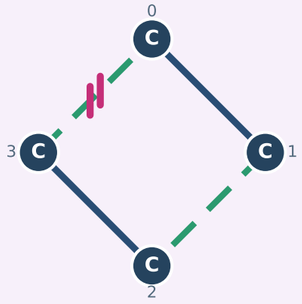
6 e⁻ • Hückel • 2π + 2π + C-C σ



suprafacial / suprafacial — **THERMAL ALLOWED**

[2+2] cycloaddition

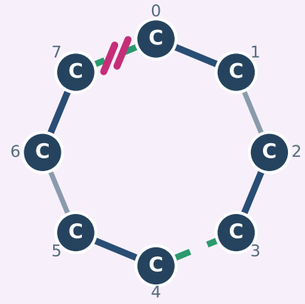
4 e⁻ • Möbius • 2π + 2π



suprafacial / antarafacial — **THERMAL ALLOWED**

[4+4] cycloaddition

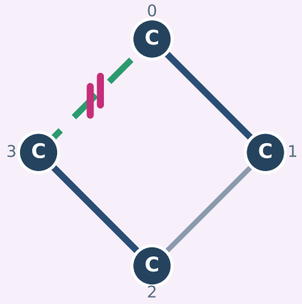
8 e⁻ • Möbius • 4π + 4π



suprafacial / antarafacial — **THERMAL ALLOWED**

4π electrocyclic closure

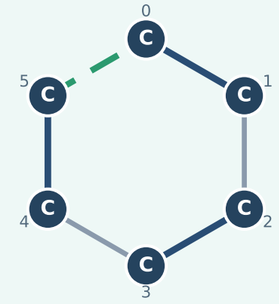
4 e⁻ • Möbius • 4π



conrotatory — **THERMAL ALLOWED**

6π electrocyclic closure

6 e⁻ • Hückel • 6π



disrotatory — **THERMAL ALLOWED**

Nodes are active orbitals (C, O, or transferring H). Each edge is an interaction in the proposed cyclic transition state.

The pink slash is one canonical phase flip; orbital sign gauges can move it. These are formal orbital topologies, not optimized 3D structures or barrier predictions.